

HARRY JIANG

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Education

University of Waterloo

September 2021 – May 2026

Bachelor of Statistics, Computer Science + Combinatorics & Optimization + Economics

Waterloo, Ontario

- Ronald G. Scoin/Rene Descartes National Scholarship (\$18,000)

Relevant Coursework: Stochastic Processes, Sampling and Experimental Design, Applied Linear Model, Computational and Mathematical Statistics, Combinatorial Optimization, Numerical Computation, Cryptography, Algorithm Design, Functional Programming, Object Oriented Programming, Data Structures and Data Management, Sequential Programming

Languages: Python, R, Java, C, C++, C#, OCaml, JavaScript, Racket, HTML/CSS, SQL, Dart

Experience

Jane Street Capital | Python, OCaml

May 2024 – August 2024

Quantitative Trader Intern

New York City

- Ongoing internship - learning about financial tools and theory, participating in frequent mock trading games, shadowing full-time traders, and working with others to build out project to be used on the trading floor.

Accutar Biotech | C++, C, Python

September 2023 – December 2023

Software Research Intern

Calgary

- Researched and developed different use-case specific compression algorithms, optimizing for compression ratio, encode/decode efficiency, and file read/write times to improve drug-protein interaction analysis by up to 30%.
- Built an imperative performance benchmarking platform to automate and visualize performance statistics, speeding up performance testing by over 80%.

Artemis Data | Python, JavaScript, HTML/CSS

January 2023 – April 2023

Software Engineer Intern

Vancouver

- Rigorously designed and implemented a novel back-end cataloging and syncing system for data organization and navigation. Reduced front-end user load times by over 99% and decreased back-end load by over 90%.
- Built up a crucial full-stack admin support application for monitoring and managing customer organization accounts, reducing new-customer onboarding time by over 75% and drastically decreasing user-support times.

Moneris Solutions | SQL, C++, Python

May 2022 – August 2022

Data Engineer Intern

Toronto

- Worked extensively with SQL databases, structuring and scripting to improve the efficiency of legacy processes.
- Helped build new methods for automating job creation and implementation, improving efficiency by over 75%.

Contest Awards

Canadian Math Olympiad/Canadian Open Math Competition

- Top 3 in BC for the COMC throughout all years of high school, qualified for and wrote the CMO 3 times.

American Invitational Math Exam/American Math Contest

- Top 5 in Canada for several AMC 10s/12s, placed top 3 in all of Canada for the AIME twice.

Canadian Chemistry Olympiad

- Earned a Gold medal (2nd in all of Canada) at the CCO and qualified for the Canadian ICHO team.

Projects

Real Estate Quantitative Analysis Tool | Python, C++

January 2023 – Present

Project Lead and Head Quantitative Researcher (4-person team)

- Used public housing data to train a multi-layered neural network consisting of a custom-trained encoder followed by a parallelized convolutional neural network (CNN) and a recurrent neural network (RNN).
- Implemented a two-phased analysis to generate better estimates for fair-market sale prices than government valuations, and to accurately create long-term growth projections for property values.

AI Traffic Light Network | Java, Python, C++

March 2019 – Present

Solo Project

- Designed and implemented dynamic traffic light algorithms built on machine learning using upstream data analysis to significantly improve traffic efficiency by decreasing time waiting at traffic lights by up to 50%.
- Collected, analyzed, and refined data by building a simulated downtown city environment and testing the algorithm against various traffic conditions, using existing algorithms as benchmarks.
- Competitive Poker, Ice Hockey, Basketball, Speedcubing, Outdoor Photography, Hiking, Cooking